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RESEARCH ARTICLE

Strategic Planning of Agroforestry Supply Chains

Thomas Kirschstein[1,2]

[1]Fraunhofer IKTS Center for Economics and Management of Materials, Halle, Germany
[2]Hochschule RheinMain – University of Applied Sciences and Arts, Wiesbaden Business School, Wiesbaden, Germany

Correspondence
Thomas Kirschstein
Email: thomas.kirschstein@ikts.fraunhofer.de

Abstract

Agroforestry systems integrate fast-growing woody perennials into agricultural land forming a biological-industrial symbiosis. They simultaneously deliver multiple biomass fractions of differing quality to food, chemical, pulp-and-paper, as well as energy industries through a shared regional supply network. The woody biomass feedstock produced in agroforestry systems can be used to substitute fossil feedstocks particularly in chemical and energy industry. Additionally, biogenic carbon can be stored in long-lived products and agroforestry increases climate resiliency and sustainability of agricultural production. In contrast to the traditional forestry-based supply of woody biomass, agroforestry-based woody biomass shows substantial feedstock potentials as most land is used agriculture in central Europe. Despite these environmental and economic potentials, agroforestry is only applied in marginal scale in central European agriculture. A core challenge in fostering agroforestry is to establish economically viable regional bioeconomy networks which connect partners from agriculture as well as downstream industries. Such an agroforestry-based supply chain needs to be structured and organized in a way that allows all partners to benefit economically in the long run. This paper presents a supply chain design model that helps to support design decisions for agroforestry-based supply chains. Therefore, it models agroforestry establishment and harvest scheduling on farm-level in a large-scaled linear multi-echelon network design model. It relies on biomass growth models which describe the production of multiple biomass types in agroforestry systems. These biomass fractions are pretreated and transported to consumers from chemical, paper-and-pulp, and energy industry depending on quality and demand restrictions. The model is applied in a case study focussing on central Germany where a heterogeneous industrial cluster is located within a fertile and large agricultural production area. The optimization results indicate that agroforestry-based supply chains can be configured in an economically viable way. Agroforestry systems can serve as feedstock suppliers for industrial consumers in the same region at a considerable scale. Profitability of agroforestry-based agricultural practice seems to be competitive with traditional agricultural production and at the same time allow producing woody biomass at a competitive cost, provided transports are not too long.

KEYWORDS
Industrial symbiosis, Agroforestry, Biomass cascading, Bioeconomy network design, Supply chain optimisation, Biobased production

1 | INTRODUCTION

The industrial bio-economy depends on connecting primary biological production systems to multiple industrial sectors through networks that efficiently allocate biomass fractions of varying quality to their highest-value application (Hildebrandt, O'Keeffe, Bezama, & Thrän, 2019). Agroforestry systems (AFS) that deliberately integrate fast-growing woody perennials into agricultural land represent realisation of this principle. By co-producing heterogeneous biomass fractions like stems, branches, twigs, leaves etc. which are more or less suitable for chemical, paper-and-pulp industry, as well as energy generation, AFS create diverse feedstocks that can serve multiple industrial consumer sectors through a shared regional supply network (Toensmeier, 2017). This biological-industrial symbiosis substitutes fossil feedstocks with renewable bio-based alternatives and stores biogenic carbon in long-lived material products (Millard, 2011).

Despite their ecological and industrial potentials, temperate AFS occupy only a marginal share of agricultural land in practice. In Germany, approximately 1,200 hectares of silvoarable AFS were established in 2025, accounting for barely 0.02% of total agricultural land (Deutscher Fachverband für Agroforstwirtschaft (DeFAF) e.V., 2026). This gap between supposed potentials and observed practices indicates implementation challenges which often root in economic, organizational, and regulatory obstacles. Establishing AFS at regional scale requires coordinated decisions on stand establishment, harvest scheduling, and biomass allocation across spatially distributed industrial consumers (Low, Dalhaus, & Meuwissen, 2023). E.g. farmers will only invest in AFS when they are convinced to generate revenues from selling wood to industrial customers at the time of harvest which is, however, 5-15 years after establishing an AFS. This long time span imposes a severe investment risk as investment cost are clear and immediate but revenue is uncertain and far away (Millard, 2011). On the other side, industrial consumers particularly in process industries like chemical and paper-and-pulp industry prefer reliable and high-volume biomass supply contracts which imposes a challenge for farmers as AFS growth depends on weather as do harvest decisions (Harayama et al., 2020). Next to harvest scheduling, a lot of other business decisions need to be made. Early harvests promise earlier revenues, but harvested biomass is dominated by small-scaled branches, twigs, and residues with limited usability in downstream industries and lower value (Kauter, Lewandowski, & Claupein, 2003). At longer rotations, biomass quantity is higher and mostly composed of stems whose diameters surpass quality thresholds such that the share of chemical- and paper-grade material rises markedly (Huber, Amann, Hülsbergen, & Schmid, 2018; Niemczyk & Thomas, 2021). The biomass feedstock quantity and quality in an AFS-based supply chain therefore depend on the rotation strategies and the industrial demand structures of multiple farms and industrial consumers. Current biomass supply chain models, reviewed in subsection 2.3, often do not consider this aspect as biomass availability is usually given at exogenous time series or optimise a single downstream supply chain rather (Zahraee, Shiwakoti, & Stasinopoulos, 2020; De Meyer, Cattrysse, & Van Orshoven, 2016).

This paper closes this gap by formulating an AFS supply chain design (AFS-SCD) problem. The linear programm integrates design and operation decisions of biomass production in AFS with multiple downstream industries via logistical processes. This way, interdependencies between biomass production decisions in AFS, logistical handling decisions, and biomass allocation decisions in down-stream processes are modeled explicitly. Biomass production decisions are based on growth models which estimate the quantity of biomass types in AFS. Thereby, biomass types are defined based on its suitability for downstream processing. Three different grades are distinguished: chemical/biorefinery grade, pulp and paper grade, and energy grade. Thereby, a quality cascade is assumed. I.e., biorefinery-grade biomass can also be used in pulp-and-paper as well as energy applications, but the reverse doesn't hold. Depending on stand age, quantities of these grades are available in AFS which guide harvest decisions. Once harvests are scheduled, biomass flows to pre-treatment and storage hubs are planned where biomass grades are allocated and transshipments to consumers are prepared depending on demand quantity and the consumers' willingness-to-pay. This way, the proposed AFS-SCD configures a supply chain for a regional AFS-based industrial ecosystem. It helps to study effects of ecological, regulatory, and economic parameters on the profitability of AFS-based supply chains. Thereby, it integrates relevant decisions of different actors from various industries all relevant in AFS-based supply chains including farmers and land-owners, wood logistics, as well as industrial down-stream sectors (like chemicals, paper or energy).

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature on AFS, biomass growth modelling, and biomass supply chain design. Section 3 presents the mathematical formulation of the AFS-SCD. Section 4 describes a case study illustrating the application of the AFS-SCD to a real-world problem. Section 5 summarizes the findings of the paper and discusses directions for future research.

2 | LITERATURE REVIEW

Agroforestry biomass systems, stand-level growth models, and biomass supply chain optimisation have each been studied extensively, but they are rarely combined in an integrated framework that links age-dependent stand dynamics to regional product-cascading value chains. This section reviews literature on

(i) agroforestry systems and economics, (ii) biomass growth models for AFS considering multiple biomass grades, and (iii) biomass supply chain design problems.

2.1 | Agroforestry systems

Agroforestry deliberately integrates woody perennials with crops and/or livestock on the same management unit, with the aim of enhancing productivity, profitability, and environmental performance (Nair, 1993; Toensmeier, 2017). In temperate climate, three distinct system types can be distinguished based on the spatial arrangement of trees and the rotation length at which woody biomass is harvested.

Short-rotation agroforestry systems (SRAFS) integrate SRC-type tree strips (typically 1–4 rows of the same fast-growing species like poplar or willow) into an otherwise arable field in an alley-cropping arrangement. Crop production continues in the inter-row strips (Huber et al., 2018). Thus, tree strips typically cover 10–30% of the total parcel area (Graves et al., 2007). Rotation lengths and total stand lifetime in SRAFS are comparable to SRCs. In SRAFS most trees are effectively edge trees, exposed to higher light and water availability than interior SRC trees. This boundary effect raises individual-tree biomass accumulation but also intensifies competition with adjacent crop rows (Huber et al., 2018).

Long-rotation agroforestry systems (LRAFS) use the same alley-cropping spatial structure as SRAFS but with rotation cycles of 12–15 years or longer (Graves et al., 2007; Nair, 1993). At this rotation length, individual trees develop substantial stem diameters and the proportion of high-value timber increases markedly, enabling premium material uses (veneer, sawlogs, engineered wood products). Therefore, also timber-quality tree species (such as walnut, cherry, oak, or high-grade poplar clones) are planted (Reisner, de Filippi, Herzog, & Palma, 2007).

SRAFS and LRAFS are not clearly differentiated in practice. First, a clear cut-off age is hard to define and, more importantly, the intended use of AFS biomass is not always clear when establishing the AFS and can also change over time. Nonetheless, when establishing AFS, the structure of AFS and its intended use after harvest affect economic viability substantially. Thus, in the present paper SRAFS and LRAFS are not considered as separate AFS systems, but the interplay of intended use, establishment, and harvesting decisions is analysed.

The decision of a landowner to establish an AFS involves weighing substantial up-front costs and long-term opportunity costs against a diversified revenue stream. On the cost side, establishment requires site preparation, planting material, and planting labour, typically amounting to 1,500–3,500 €/ha for SRC or SRAFS (Faasch & Patenaude, 2012; Testa, Di Trapani, Foderà, Sgroi, & Tudisca, 2014). Recurrent harvest and refitting costs (machine hire, chipping, transport to roadside) add approximately 200–400 €/ha per rotation depending on stand density and equipment availability (Testa et al., 2014; Spinelli, Nati, & Magagnotti, 2009). During the rotation, maintenance costs of AFS arise on a yearly basis. Next to expenses for establishment, maintenance and harvesting, opportunity costs arise as no revenues from crops are generated on tree strips. In regions with fertile soils and high cereal prices, yearly crop revenues may sum up to 400–600 €/ha (Faasch & Patenaude, 2012; Graves et al., 2007). Thus, opportunity costs are often the dominant cost driver in SRAFS profitability analyses.

On the benefit side, the primary revenue of AFS is the sale of harvested woody biomass, whose magnitude and frequency depends on rotation length, species, and site productivity. For poplar-based SRAFS in Central Europe, typical yields range from 10 to 25 t of dry matter biomass per ha for typical rotation lengths (Huber et al., 2018; Niemczyk & Thomas, 2021). Additionally, positive crop-yield effects in the inter-row strips of SRAFS and LRAFS may result. Tree rows act as windbreaks, reducing soil erosion and evapotranspiration. They provide partial shading that moderates heat stress during drought years and their root systems may improve water retention in top-level soil (Graves et al., 2007; Grünewald, Böcker, Schneider, Bens, & Hüttl, 2007). Empirical measurements in Central European alley-cropping trials indicate that these microclimate effects can increase cereal yields in adjacent strips by 5–15% relative to open-field controls, partially or fully compensating the loss of arable area occupied by the tree strips (Graves et al., 2007). Beyond direct yield effects, AFS deliver broader ecosystem services (like soil organic matter accumulation, carbon sequestration, biodiversity enhancement) that may lead to agri-environment subsidies and carbon credits which have to be considered as additional financial benefits (Toensmeier, 2017; Faasch & Patenaude, 2012).

Nonetheless, economic attractiveness of AFS depends primarily on revenues of AFS biomass. Thereby, revenues are different for the types of biomass and in which value chain it is used. Often AFS biomasses are used energetically, i.e. for heat and power generation, or anaerobic digestion, which typically creates low prices per tonne of biomass (Keegan, Kretschmer, Elbersen, & Panoutsou, 2013). Alternatively, woody parts of the biomass (i.e., the stem and large branches) can also be used as pulp and paper grade material as well as feedstock for chemical industry or as timber achieving higher prices (Lamers, Junginger, Hamelinck, & Faaij, 2011). This price differentiation mirrors long-established forestry product classification systems, in which round timber is sorted into quality classes. These classes range from veneer logs and sawlogs at the top, through pulpwood, to fuelwood and energy wood at the bottom (Sandberg et al., 2023). Wood categorization thereby depends primarily on stem diameter, straightness, and the presence of defects such as knots or curvature (Petráš, Mecko, & Nociar, 2008). Because larger, defect-free stems can be directed to higher-value sawlog and veneer markets while smaller-diameter or lower-quality material is confined to pulp, particleboard, or energy uses, this classification links AFS stand age and management directly to achievable biomass revenues. SRAFS typically supply the lower end of this quality cascade, LRAFS allow producing larger-diameter, higher-grade stems (Skog, Abt, & Abt, 2014). Thus, fostering AFS establishment by maximizing economic returns from AFS biomass requires a strategy for allocating biomass fractions to the highest-value applications, while ensuring that the rotation length is aligned with the growth dynamics of the stand and the market demand for different biomass types (Lamers et al., 2011).

Thus, for making educated decisions on AFS establishment and harvesting, the expected biomass quantities at harvest for stem, branches, and residues need to be estimated. Thereby, the shares of the biomass types change with stand age. E.g.~the share of stem weight rises from roughly 55% at age 4 to over 75% at age 10 (Huber et al., 2018; Testa et al., 2014). Therefore, in subsection 2.2 growth modelling approaches are reviewed that capture the age-dependent growth dynamics of different biomass types in AFS.

2.2 | Biomass growth models

Reliable estimation of stand-level biomass quantities and their age-dependent composition is a necessary prerequisite for strategic analysis of AFS-based supply chains. Individual-tree biomass is commonly approximated by species-specific allometric power laws relating stem base diameter to tree biomass. Such relationships have been calibrated e.g. for Central European poplar clones under SRC and SRAFS conditions (Huber, Hülsbergen, & Schmid, 2016; Huber et al., 2018). For supply chain planning, however, the relevant quantity is stand-level biomass per hectare over a multi-decade rotation horizon, which requires a temporal growth model. In the following, temporal biomass growth models for different biomass fractions are derived. Therefore, a set of variables is used which are summarized in Table 1 in the appendix.

Stand-level biomass per hectare is denoted by $M(t)$. Thereby, stand-level biomass growth can be described by different growth functions like the logistic, Gompertz, von Bertallanfy, and Richards models which are also applied for a multitude of other growth processes in biology (E. Tjørve & Tjørve, 2010). All these functions show a right-skewed sigmoidal pattern and their shape is controlled by various parameters (E. Tjørve & Tjørve, 2010). Thereby more parameters usually imply a more flexible growth model, but more parameters require more calibration efforts which have to be counterweighted with modelling accuracy (Diéguez-Aranda, Castedo-Dorado, Álvarez-González, & Rojo, 2006). Empirical studies indicate that most growth models can be calibrated to empirical observations quite accurately (Barrio-Anta, Sixto-Blanco, Cañellas-Rey de Viñas, & Castedo-Dorado, 2008). Therefore, in the following, a standard 3-parameter Gompertz model (K. M. Tjørve & Tjørve, 2017) is chosen for modelling biomass quantity, which is defined as

$$M(t) \;=\; A(N) \cdot \exp\big(-\exp(-k \cdot (t - t_0))\big), \tag{1}$$

where t_0 is the age of maximum instantaneous biomass increment and k determines how quickly the stand approaches its site-specific asymptote $A(N)$, i.e. the maximum stand-level biomass quantity for $t \to \infty$. For Central European poplar under medium-rotation AFS management (10–20 yr), analysing empirical data from (Huber et al., 2018),(Niemczyk & Thomas, 2021), and (Jha, 2018) yields parameter intervals of $k \in [0.17, 0.2]$ and $t_0 \in [9, 10]$ years, depending on local specifications of the AFS under study. Parameter t_0 is empirically estimated by analyzing current annual increments (CAI) and indicates an orientation for the optimal rotation age as growth slows for higher ages.

The asymptote $A(N)$ biomass quantity depends on multiple factors, particularly on planting density and soil quality. It can be modelled by a competition-density power law (Kira, Shinozaki, & Hozumi, 1961):

$$A(N) \;=\; C_{\text{site}} \cdot N^{1-\beta}, \tag{2}$$

where C_{site} encodes site quality indicator, N indicates tree density (e.g. trees per hectare), and β is the competition exponent. Equivalently, the per-tree asymptote $A_{\text{tree}}(N) = A(N)/N = C_{\text{site}} \cdot N^{-\beta}$ decreases with density as individual trees have access to less growing space and more nutrition competition. Parameters C_{site} and β are highly site- and species-specific. For Poplar clones under European conditions, $\beta \in [0.3, 0.5]$ and $C_{\text{site}} \in [4, 6.5]$ (Niemczyk & Thomas, 2021). Figure 2 in the appendix shows both asymptote functions according for advantageous and disadvantageous site conditions.

When assessing potential downstream application of AFS biomass, biomass fractions are to be defined meeting the requirements of downstream consumers (?, ?). E.g. for timber and biochemical applications, stem diameters of AFS trees need to exceed a minimum threshold in order to qualify as a suitable feedstock (Zhu, 2011). In the following, three biomass fractions are distinguished based on common forestry standards: stems, branch wood, and residue. The stem fraction consists of stems with a diameter of at least 15 cm, while branch wood requires a diameter of at least 7 cm. The residue fraction comprises smaller parts of the tree. To subdivide the total AFS biomass modelled by (1) into the three fractions depending on tree age, a logistic growth model (Zeide, 1993) is used:

$$q_p(t) \;=\; \frac{f_p}{1 + e^{-r_p \cdot (t - t_{50,p})}}, \tag{3}$$

where f_p denotes the asymptotic share of fraction p for mature trees. Parameter $t_{50,p}$ indicates the inflection point and coincides with the age at which 50% of the asymptotic share of fraction p is reached. Using empirical data from (Civitarese, Acampora, Sperandio, Assirelli, & Picchio, 2019), (Jha, 2018), and (Niemczyk & Thomas, 2021), $r_{stem} = 0.467{,}\text{yr}^{-1}$, $t_{50,stem} = 8.19$ years, and $f_{stem} = 0.75$ is estimated for the stem fraction. For the branch fraction, parameters $r_{bran} = 0.698\ \text{yr}^{-1}$, $t_{50,bran} = 7.55{,}\text{yr}$, and $f_{bran} = 0.15$ can be assumed. Figure 1 illustrates the development of the fraction shares over stand age. The share of the residue fraction is implicitly defined as $q_{resid}(t) = 1 - q_{stem}(t) - q_{branch}(t)$.

Combining equations (1)–(3), the dry biomass yield of fraction p per hectare at harvest age t is:

$$\eta_p(t) \;=\; q_p(t) \cdot M(t). \tag{4}$$

Fresh-weight yields at felling are obtained from dry matter yields via $\eta_p^{\text{fw}}(t) = \eta_p(t)/(1 - \omega)$, where the moisture content at felling $\omega = 0.55$ (Civitarese et al., 2019). Figure 1 shows the fractions' stacked biomass quantities $\eta_p(t)$ depending on stand age compared with empirical observations from (Jha, 2018) and (Huber et al., 2018).

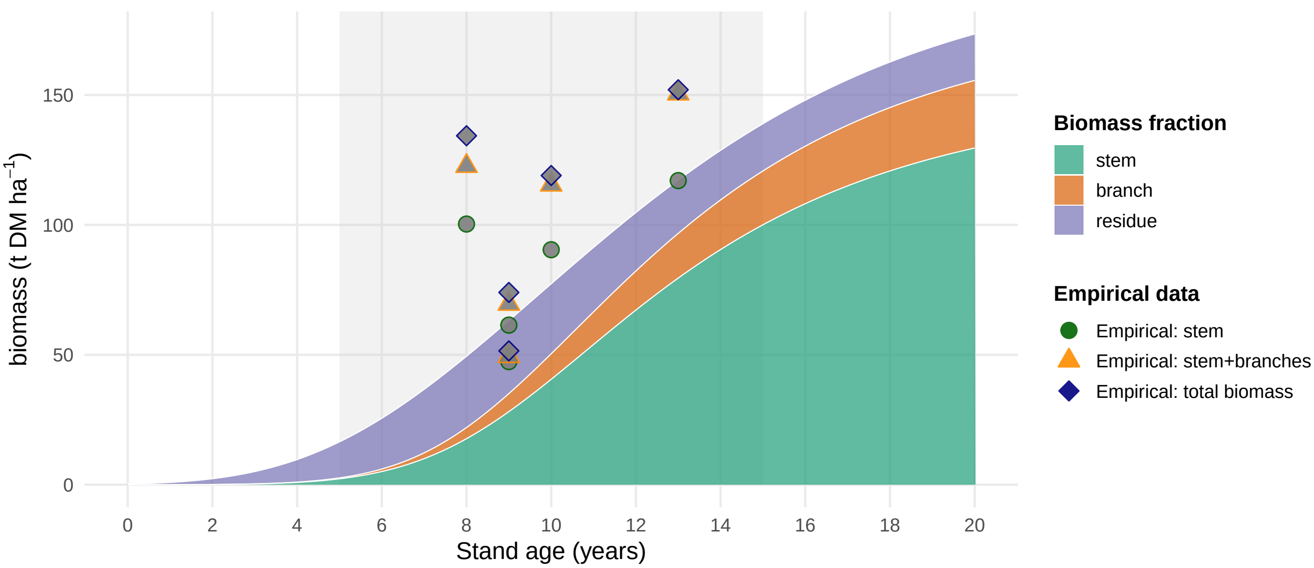


FIGURE 1 Stand-level aboveground biomass components for poplar under AFS management with $N = 250$ trees, ha^{-1}. Grey band: typical rotation times (5-15 years).

2.3 Biomass supply chain design

The design and planning of supply chains based on industrial biomass feedstocks has attracted substantial research attention over the past decades, driven by the need to ensure cost-efficient and sustainable feedstock provision for bioenergy and biorefinery applications (see e.g. extensive reviews from (Gital & Bilgen, 2024), (Zahraee et al., 2020), and (De Meyer, Cattrysse, Rasinmäki, & Van Orshoven, 2014)). Despite its name, the term biomass supply chain (BSC) is often associated with exclusive energetic uses of biomass and often equivalent to biorefinery supply chain (see e.g. (Espinoza, Bautista, Narvaez, Alfaro, & Camargo, 2017)). The considered biomass feedstocks range from agricultural energy crops, over urban biomass residues to forestry wood (see e.g. (Sharma, Ingalls, Jones, & Khanchi, 2013) for an overview). Thereby, literature on forestry supply chains (FSC) is the oldest sub-branch in this literature stream (Cambero & Sowlati, 2014). Here, the focus is on one particular biomass feedstock, but usually multiple fields of applications are considered (see e.g. (D'Amours, Rönnqvist, & Weintraub, 2008) for an overview of fields of applications). In general, the structure of BSC and FSC typically encompasses three process types: i) biomass production, ii) harvesting and logistics, and iii) biomass processing. These processes are modeled at various levels of details depending on the scope of decisions to be supported. The decision scopes can be distinguished in: (i) strategic network design (e.g. facility location, tree selection), (ii) tactical planning (e.g. harvest scheduling, inventory capacities), and (iii) operations (e.g. routing of vehicles, inventory control) (De Meyer et al., 2014). This manuscript is dedicated to agroforestry supply chain design. I.e. the focus is put on a specific type of biomass and multiple fields of application. In the following relevant literature from FSC and BSC literature is reviewed which contributes to this type of problem. Relevant aspects from BSC literature mainly involve biomass production economics and strategic design modelling while relevant aspects from FSC literature is concentrated on wood logistics and industrial applications. Note that there is no existing literature dedicated on agroforestry supply chain design to the best of the author's knowledge.

One of the earliest FSC models is proposed by (Gunnarsson, Rönnqvist, & Lundgren, 2004). Here, a cost-optimal distribution and processing network for supplying heating plants with wood chips is to be found. Therefore, a multi-period MILP model is formulated satisfying feedstock demand at heating plants by selecting harvesting areas and deciding in each period (i.e., each year) about chipping, storing, and transporting forest biomass (like logs). (Eksioglu, Acharya, Leightley, & Arora, 2009) develop one of the earliest MILP models for biomass-to-biorefinery supply chain design. The focus is on biomass conversion processes by integrating facility location, capacity sizing, and transport mode selection for lignocellulosic feedstocks. The model minimises total system cost assuming, that biomass supply is an exogenous input, i.e. biomass production is not modelled explicitly. (De Meyer et al., 2016) proposes a similar profit-maximising MILP for strategic and tactical decision support. In this model, the focus is on pre-treatment and storage processes of different types of grassland biomass types (grass, hay, straw, etc.) supplying anaerobic digesters for energy generation. (Walther, Schatka, & Spengler, 2012) describe wood-based biomass supply chain for biofuel production where the biomass is converted either directly into biofuels or indirectly via an intermediate material (slurry). The objective is here maximize the net present value (NPV) over the planning horizon discounting the yearly cash flows of the supply chain. The revenue is created by supplying refineries with the supply chain's biofuels. Similar to (Gunnarsson et al., 2004), the refineries' biofuel demand can be satisfied by imports, such that a possible solution is that no regional biomass is used to produce biofuel. The aforementioned models consider only biomass-to-energy supply chains.

(Mansoornejad, Pistikopoulos, & Stuart, 2013) extend this perspective by integrating product and process portfolios. Their model considers multiple biomass conversion processes as well as sources of lignocellulosic biomass. It accounts for product conversion efficiencies along different processing pathways and maximises profit over a multi-period time horizon. Likewise, (Cambero, Sowlati, Marinescu, & Röser, 2015) model energetic and chemical conversion pathways of lignocellulosic biomass in a strategic supply chain design MILP for maximizing NPV over a 20-year period. (García-Velásquez, Leduc, & van der Meer, 2022) propose a bi-objective MILP for designing a supply chain for producing Polyethylene Terephthalate (PET) based on wheat and sugar beets. Thereby, both biomass types are required to produce intermediate chemicals (mono ethylene glycol and purified terephthalic acid) which are merged into PET such that a 4-stage supply chain is modeled. For each processing step locations have to be selected among a set of candidate locations. The objectives pursued involve total supply chain costs and GHG emissions. The trade-off between cost and GHG emissions is studied by incorporating GHG cost and varying GHG prices. A similar 4-stage bio-chemical network design problem for biobutanol production based on microalgae is studied in (Arabi, Yaghoubi, & Tajik, 2019). Recently, (Roudneshin & Sosa, 2025) presents a novel procedure for a bioenergy supply chain design problem by combining mathematical programming with an Analytic Hierarchy Process (AHP). The AHP pre-processes potential candidate locations for biorefineries where biomass wastes are converted into biogas which is subsequently fed into the local gas grid.

All BSC and FSC literature reviewed above does not consider biomass production processes explicitly, but model biomass as an exogenous resource. This is because most approaches concentrate on application where waste biomass is considered as feedstock. Also in FSC applications production decisions are usually not part of supply chain design decisions as growing cycles in forest management are even longer than the time horizons in supply chain design problems. This manuscript contributes to literature by proposing a novel agroforestry supply chain design problem (AFS-SCD) which explicitly models biomass production as an integral part of the decision support. In agroforestry applications, biomass production decisions depend on the existence of downstream markets for the produced wood. Moreover, multiple types of biomass and multiple application industries are considered. In the following the AFS-SCD is outlined in detail.

3 | MILP MODEL FOR AGROFORESTRY SUPPLY CHAIN DESIGN

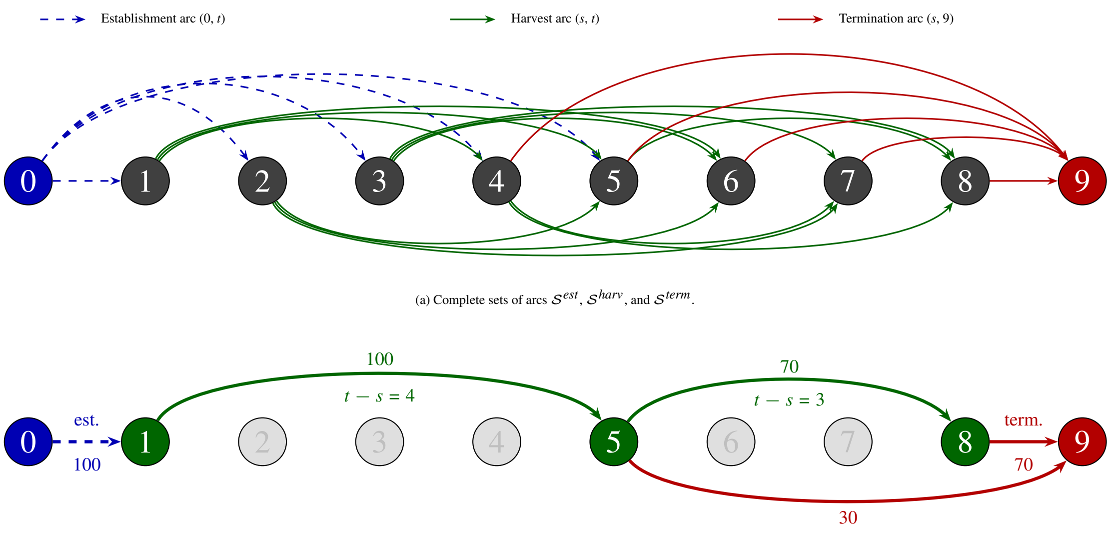
Building on the biomass growth models introduced in subsection 2.2 and the supply chain design literature reviewed in subsection 2.3, this section describes the AFS supply chain design problem (AFS-SCD), a mixed-integer linear programming (MILP) model for the integrated design and operation of an agroforestry biomass supply chain with multiple AFS biomass types (i.e., stem, branch, and residue fractions) used as feedstocks for potential consumers in chemical, pulp and paper as well as energy industry.

3.1 | Sets, Indices, and Time Structure

The model considers a set of agroforestry sites $\mathcal{I}$, storage/pre-treatment facilities $\mathcal{J}$, consumer sites $\mathcal{K}$, and biomass types $\mathcal{P} = \{stem, branches, residues\}$, whose growth models are outlined above. Time is discretised into annual periods $t \in \mathcal{T} = \{1, \ldots, T_{\max}\}$. To reduce complexity, reflect legal definitions of SRAFS and LRAFS as well as discard implausible options (e.g. extremly long rotation cycles), $A^{\min}$ and $A^{\max}$ define minimum and maximum stand ages. Thus, harvest are possible in periods $\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \ldots, T_{\max}\}$. To consistently model establishment and consecutive harvest decisions, the following arc sets are defined: $\mathcal{S}^{est}$ defines potential establishments of AFS. If an AFS is established in period $t \in \mathcal{T}$, arc $(0, t)$ is selected. Harvest set $\mathcal{S}^{harv}$ summarizes all potential consecutive harvest periods within the age interval $[A^{\min}, A^{\max}]$. I.e., if an AFS is established or harvested in period s and next harvest is in period t, arc (s, t) is selected. Finally, if there is no subsequent harvest after period t, the AFS is terminated by selecting arc $(t, T^{\max} + 1)$. Termination arcs are summarized in set $\mathcal{S}^{term}$. All arc sets are formally defined as follows:

$$\begin{aligned}
\mathcal{S}^{est} &= \{(0,t) \mid t \in \{1,\ldots,T^{\max} - A^{\min}\}\},\\
\mathcal{S}^{harv} &= \{(s,t) \mid s,t \in \mathcal{T} \wedge t - s \in \{A^{\min},\ldots,A^{\max}\}\},\\
\mathcal{S}^{term} &= \{(s,T^{\max}+1) \mid s \in \{A^{\min}+1,\ldots,T_{\max}\}\},\\
\mathcal{S} &= \mathcal{S}^{est} \cup \mathcal{S}^{harv} \cup \mathcal{S}^{term}.
\end{aligned}$$

Figure 2a illustrates this network structure of all arc sets for $T = 8$, $A^{\min} = 3$, $A^{\max} = 5$.



(a) Complete sets of arcs $\mathcal{S}^{est}$, $\mathcal{S}^{harv}$, and $\mathcal{S}^{term}$.

(b) Example of a feasible path through the network with establishment in period 1, harvests in periods 5 and 8, termination in period 9.

FIGURE 2 Network structure of set $\mathcal{S}$ for $T = 8$, $A_{\min} = 3$, $A_{\max} = 5$. Dashed blue arcs: establishment arcs $\mathcal{S}^{est}$, solid green arcs: harvest arcs $\mathcal{S}^{harv}$, solid red arcs: termination arcs $\mathcal{S}^{term}$.

3.2 | Decision Variables

Temporal flow Variables z_{ist} indicate on which quantity of the area of site i an an AFS is operated between $(s,t) \in \mathcal{S}$. Depending on the arc type, this implies establishment, harvest, or termination. Selected arcs build a network flow over the complete planning horizon from $t = 0$ to $T^{max} + 1$. Figure 2b illustrates a simple network flow with establishment of an AFS in period 1 and harvests in periods 5 and 8.

Spatial flow variables X_{ijpt} and $X_{jkpp't}$ represent transport of products from AFS sites to storage/pre-treatment facilities and from there to consumers, respectively, while S_{jpt} captures end-of-period inventory at facility j. Table 2 in the Appendix summarises all sets, decision variables, and parameters used in the MILP formulation, while η_{pa} refers to (4).

3.3 | Objective Function

As discussed in Section subsection 2.1, the economic attractiveness of AFS supply chains is modelled from a global perspective taking into account all processes, activities, and participants involved in finally serving customer markets with AFS-based feedstocks. Therefore, revenues for the different biomass fractions and are weighed against costs occurring at all value-adding stages of the supply chains. Revenues created by selling AFS-based feedstocks to different industrial consumers are counterweighted by establishment, opportunity, harvest, transport, and storage costs of landowners, agricultural firms as well as biomass logistics companies. The model therefore maximizes total profit of all participants in the AFS supply chain over the planning horizon as follows:

$$\begin{aligned}
\max\ Z = & \sum_{k\in\mathcal{K}}\sum_{p\in\mathcal{P}} R_{kp}\cdot\sum_{j\in\mathcal{J},}\sum_{(p',p)\in\mathcal{Q}}\sum_{t\in\mathcal{T}} X_{jkp'pt} \\
& -\sum_{i\in\mathcal{I}} c_i^{est}\cdot\sum_{(0,t)\in\mathcal{S}^{est}} z_{i0t} \\
& -\sum_{i\in\mathcal{I}}(c_i^{main}+c_i^{opp})\cdot(t-s)\cdot\sum_{(s,t)\in\mathcal{S}^{harv}} z_{ist} \\
& -\sum_{i\in\mathcal{I}} c_i^{harv}\cdot\sum_{(s,t)\in\mathcal{S}^{harv}} z_{ist} \\
& -\sum_{i\in\mathcal{I}}\sum_{j\in\mathcal{J}}\sum_{p\in\mathcal{P}}\sum_{t\in\mathcal{T}} c_p^{\text{tr-raw}}\cdot d_{ij}\cdot X_{ijpt} \\
& -\sum_{j\in\mathcal{J}}\sum_{k\in\mathcal{K}}\sum_{(p,p')\in\mathcal{Q}}\sum_{t\in\mathcal{T}} c_{p'}^{\text{tr-pre}}\cdot d_{jk}\cdot X_{jkpp't} \\
& -\sum_{j\in\mathcal{J}}\sum_{p\in\mathcal{P}}\sum_{t\in\mathcal{T}} c_j^{stor}\cdot S_{jpt}
\end{aligned} \tag{5}$$

Here, R_{kp} are product- and consumer-specific revenues for each ton of biomass fraction (€/t). The opportunity cost term penalises the land use when AFS are established. Thereby, c_i^{opp} can be negative or positive depending on the site-specific conditions. It comprises effect of lost field crop revenues due to AFS as well as higher yields of remaining field crop areas e.g. due to wind and water erosion protection by AFS. While the assessment of lost field crop revenues is usually well predictable, the economic evaluation of positive ecological effects of AFS (e.g. on neighboring crop yields) is more difficult. Thus, c_i^{opp} can also contain regulatory subsidies to support establishing and maintaining AFS. Transport cost terms mirror the two-stage flow structure of the AFS supply chain with differentiated rates for raw and pre-treated biomass depending on the biomass type. Cost rates differ because as e.g. the density and water content of the shipped biomass types differ or because specialized equipment (e.g. for log transports) is used (Väätäinen et al., 2021). Moreover, site-specific establishment cost for planting the AFS trees is considered, next to yearly maintenance and harvesting costs. All these costs scale with the size of a site ($AREA_i$) measured in hectares. Maintenance cost comprise e.g. costs for trimming, replanting, and pruning unnecessary shoots. Storage costs are considered if biomass types are stored for more than a year. Storage cost rate c_j^{stor} measures the costs for storing a ton of biomass for one year and comprises direct storage cost (e.g. for storage site maintenance) as well as opportunity cost due to potential material losses and capital commitment. Biomass types harvested and consumed in the same year incur no storage cost, as usually AFS are harvested at the beginning of a year. Harvested biomass has to dry during spring and summer such that the biomass can be used in autumn or winter the same year. As under-year drying is necessary for some industrial applications and beneficial for transport economics, drying storage in the year of harvest is not associated with additional cost in this study.

3.4 | Constraints

Constraints (6) limits the area dedicated to AFS operations to the maximum size of site i. Constraints (7) enforce flow conservation in each period. If AFS systems of size z_{ist} are established or harvested in period $t \in \mathcal{T}$, there needs to be a succeeding activity (harvest or termination) of this area. This way, (7) guarantee valid (possibly empty) AFS activity over the complete time horizon for each site. Constraints (8) link the harvest-cycle variables z_{ist} to physical yield quantities $\eta_{p(t-s)}^{fw}$ for age $t-s$, which are precomputed using (1)–(4) scaled by moisture content ω. Thus, constant yields per hectare are assumed for each site i. Note that this simplification can be dropped and site-specific yields $\eta_{ip(t-s)}^{fw}$ could be introduced without increasing numerical complexity of the AFS-SCD. Moreover, harvest quantities can be smaller than available AFS biomass implying that unharvested biomass quantities are lost. Constraints (9) link harvest quantities to transport flows and ensure inventory balance at each facility. Thereby, biomass losses due to drying ($\Delta \leq 1$) and biomass conversion processes ($\delta_{pp'} \geq 1$) are considered. Constraints (10)–(11) enforce storage and processing capacity limits at each facility j. Constraints (12) implement product usage at multiple consumption sites k. Thereby, $(p',p) \in \mathcal{Q}$ defines a product mapping allowing high-quality product p' (e.g. stems) satisfying demand of a lower-quality products p up to maximum demand $D^{\max}_{kpt}$.

$$\sum_{(0,t)\in\mathcal{S}^{est}} z_{i0t} \leq AREA_i \qquad \forall\, i \in \mathcal{I} \tag{6}$$

$$\sum_{(s,t)\in\mathcal{S}} z_{ist} = \sum_{(t,u)\in\mathcal{S}} z_{itu} \qquad \forall\, i \in \mathcal{I},\ t \in \mathcal{T} \tag{7}$$

$$\sum_{j\in\mathcal{J}} X_{ijpt} \leq \sum_{(s,t)\in\mathcal{S}^{harv}} \eta^{fw}_{p(t-s)} \cdot z_{ist} \qquad \forall\, i \in \mathcal{I},\ p \in \mathcal{P},\ t \in \mathcal{T}^{harv} \tag{8}$$

$$S_{jpt} = \Delta \cdot S_{jp,t-1} + \sum_{i\in\mathcal{I}} X_{ijpt} - \sum_{k\in\mathcal{K},\,(p,p')\in\mathcal{Q}} \delta_{pp'} \cdot X_{jkpp't} \qquad \forall\, j \in \mathcal{J},\ p \in \mathcal{P},\ t \in \mathcal{T}^{harv} \tag{9}$$

$$\sum_{p\in\mathcal{P}} S_{jpt} \leq \mathrm{CAP}^{stor}_j \qquad \forall\, j \in \mathcal{J},\ t \in \mathcal{T} \tag{10}$$

$$\sum_{i\in\mathcal{I},\,p\in\mathcal{P}} X_{ijpt} \leq \mathrm{CAP}^{proc}_j \qquad \forall\, j \in \mathcal{J},\ t \in \mathcal{T}^{harv} \tag{11}$$

$$\sum_{j\in\mathcal{J},\,(p',p)\in\mathcal{Q}} X_{jkp'pt} \leq D^{\max}_{kpt} \qquad \forall\, k \in \mathcal{K},\ p \in \mathcal{P},\ t \in \mathcal{T}^{harv} \tag{12}$$

$$z_{ist},\ X_{ijpt}, X_{jkp'pt},\ S_{jpt} \geq 0 \qquad \forall\, i,j,k,p,p',t \tag{13}$$

In summary, AFS-SCD constitutes a large-scale LP which allows to decide on which fraction of a site, AFS are established and how the AFS operations are scheduled in a multi-year planning horizon. The AFS-SCD takes a supply chain perspective and maximizes total profit of all actors involved in these processes. Thus, it does not explain how the profits are split e.g. between farmers, AFS service providers and logistics service providers. The continuous modelling of AFS areas in z_{ist} leads to a comparatively easy to solve problem. On the other side, it is unclear which part of a site's total area is chosen for AFS operations if $z_{ist} < AREA_i$, which holds for establishment as well as harvesting and termination.

4 | CASE STUDY

The AFS-SCD is intended to support the design of regional agroforestry biomass supply chains fostering regional bioeconomy clusters thereby connecting multiple industrial sectors. The proposed AFS-SCD model is instantiated on a real-world biomass supply chain case study located in the tri-state region of Saxony-Anhalt, northern Thuringia, and northern Saxony, Germany. This region is characterized by a high density of agricultural land usage as well as cluster of paper and wood as well as chemical industry. The regional chemical industry relies on fossil resources so far, but is under substantial transformative pressure which led to investments in sustainable production processes requiring sustainable regional bio-based feedstocks. Figure 3 in the Appendix shows agricultural production sites along with the set of potential wood-processing hubs, and the regional users of AFS biomasses from chemical, paper and wood processing industry as well as the energy sector.

The supply chain comprises 2239 agricultural production parcels which are grouped into $|\mathcal{I}| = 145$ potential AFS production sites. Clustering is based on geographical distances between parcel centroid using a k-means approach in order to reduce problem size. For distance calculations, for each site a centroid is calculated. Pre-treatment and intermediate storage facilities $\mathcal{J}$ are located on existing wood processing facilities as well as potential new log or wood chip yards. All harvested biomass quantities at AFS sites are transported to logistics hubs first. At the hubs sorting and transshipment processes take place before biomass types are forwarded to the consumer facilities. Storing and transshipment capacities at the hibs are assumed to be constant in time and universally set to 4.0909×10^4 kt per year. Finally, $|\mathcal{K}| = 28$ consumer sites are included. Each consumer site belongs to an existing facility from chemical, paper/wood processing industry, or the energy sector. Distances between the facilities d_{ij} and d_{jk} are calculated as road distances between the facilities' coordinates. For each consumer facility, an yearly upper demand capacity for each of the three distinguished biomass product types are given along with consumer-specific prices. The following biomass types are distinguished:

- **P1: stem wood – chemical/paper grade**: long-fibre wood chips from stems suitable for conversion in a local biorefinery and paper factory. In total, a yearly demand of 620 kt and an average prices of 45 € is assumed.
- **P2: branch wood – sawmill/pulp grade**: medium/short-fibre wood chips from larger branches suitable for mechanical processing in sawmills and lower-grade pulping. Demand is distributed across 6 smaller facilities and accumulates to 410 kt/year. Gate prices range from 28-38 €/t.
- **P3: residue – energy grade**: residual wood fractions (chips, bark, off-cuts) used for heat and electricity generation or lower-grade pellet production. Demand is accumulates among 24 consumers to 132 kt/year. Gate prices range from 18-24 €/t.

Table 3 in the appendix summarizes the consumer-specific demand data. It is assumed that the biomass product types show a quality cascade which means that the demand of pulp grade (P2) can also be satisfied by chemical/paper grade biomass (P1). Likewise, energy grade demand (P3) can be satisfied by all biomass types. Note that constant demands over the planning horizon are assumed, although an increasing biomass demand can be expected due to the green transition in those industries. From this perspective, a rather conservative market demand scenario is assumed.

Growth model introduced in subsection 2.2 is used to model biomass quantities depending on age and the area dedicated to AFS. The growth model is parameterized for poplar clones at central European conditions as introduced above with a planting density of $N = 400$ trees per hectare. If AFS are established, they have to be harvested within $A^{min} = 3$ to $A^{max} = 20$ years allowing a wide range of potential harvesting strategies between short-rotation and long-rotation AFS. As the total planning horizon is $|\mathcal{T}| = 40$ years multiple harvesting cycles can be established. The sites do not differ with respect to growth conditions as the climatic and soil quality variation within the comparatively small region is rather limited. However, the sites differ with respect to size (given in hectares) and opportunity cost rates (given in € per hectare and year). Site-specific opportunity cost rates c_i^{opp} is normally distributed with mean 521 and standard deviation 391. For most sites, substantial opportunity costs are assumed reflecting the loss in field-crop revenue, risk premiums of landowners as well as higher costs for field crop maintenance. The 145 sites with negative opportunity cost rates reflect sites where AFS lead to an increase in field crop productivity on the remaining field area. The variation in opportunity cost rates to study the effect of opportunity cost on site selection for AFS and gives valuable insights particularly for communication with farmers and politics. Subsidies for establishing AFS are one of the core instruments to foster AFS. The height and effectiveness of those subsidies, however, is controversially disputed. Remaining cost parameters are universally set for all sites and comprise maintenance cost rates c_i^{main} (€ per year and hectare), establishment cost c_i^{est} (€ per hectare) and harvesting cost c_i^{harv} (€ per hectare). Likewise, transport cost rates for harvested biomasses c_p^{tr-raw} and pre-treated/sorted biomasses $c_{p'}^{tr-raw}$ are set universally. All scalar parameters are shown in Table 4 in the Appendix.

4.1 | Optimization results

The AFS-SCD model is solved for the instance data described above with Gurobi. The model selects a subset of sites for AFS establishment and identifies the optimal harvest schedules, intermediate facility utilisation, and product allocation across the planning horizon.

Figure 3 shows the spatial configuration of the profit-optimal supply chain.

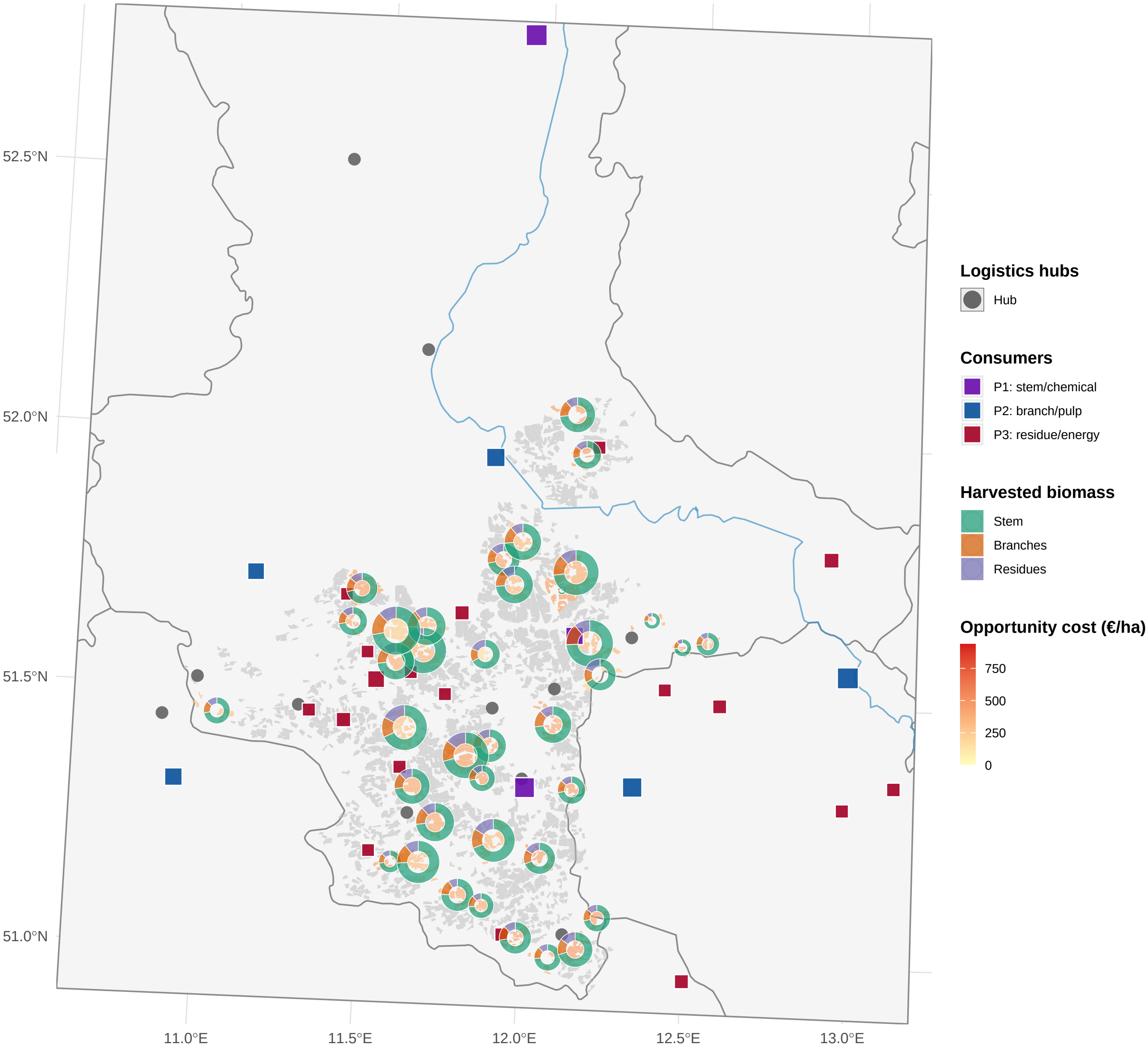


FIGURE 3 Site selection and biomass production in optimal supply chain configuration. Green polygons: active sites with biomass harvest; grey polygons: non-selected sites. Pie charts at active sites show the product mix of harvested biomass, scaled by total harvested volume.

In total, on 38 sites AFS are established which corresponds to about 26 % of all sites and 24 % of the total farm area. Over the planning horizon, 2.6368×10^4 kt of AFS biomasses are produced on these sites which suffices to satisfy 57 % of total demand. Figure 4 shows the harvested biomasses over time along with total demand of all consumer sites.

During the first 10 years no AFS biomasses are harvested as trees of established AFS in first years grow up to a certain size before first harvest. Thus, about 25% of consumers' total biomass demand cannot be

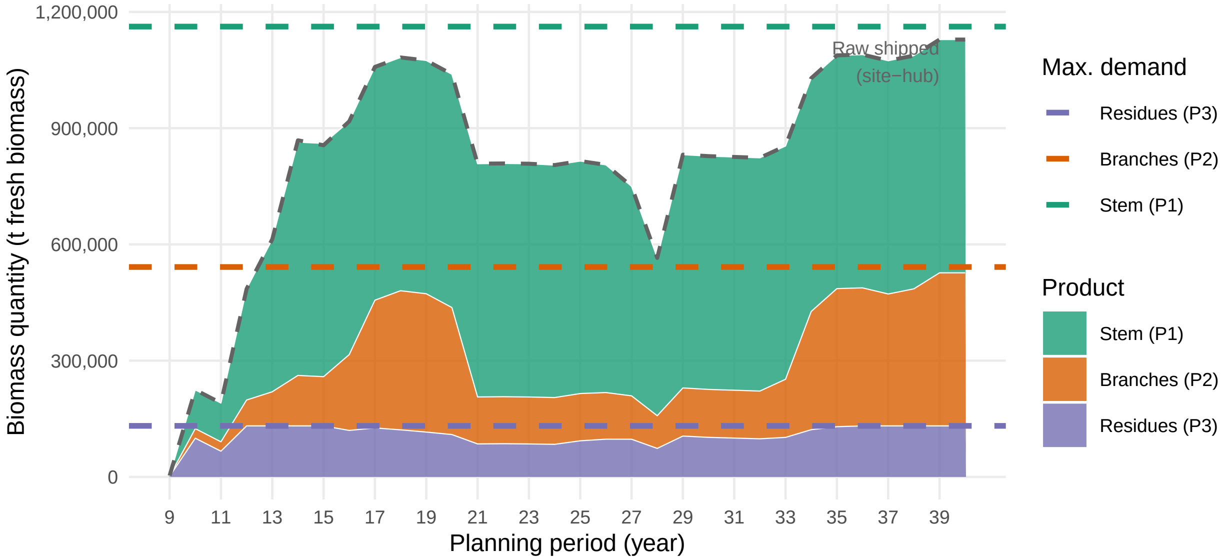


FIGURE 4 Total biomass delivered to consumers by product type over the planning horizon. Stacked areas show period-level volumes of stem (P1), branches (P2), and residues (P3) flowing from storage hubs to all consumers. Dashed line: total harvested raw biomass shipped from sites to hubs.

satisfied due to this grow-up period. Afterwards, a cyclical pattern appears with a harvesting peak around year 20 and a breakdown in year 28 before plateau is reached from year 35-40. Thereby, the harvesting quantities of stems and branches show a similar pattern while quantities of harvested energy grade biomass are rather stable and close to the demand limit over the planning horizon. Thus, rather long-rotation AFS are established as Figure 5 confirms showing the histogram of harvesting ages across all sites.

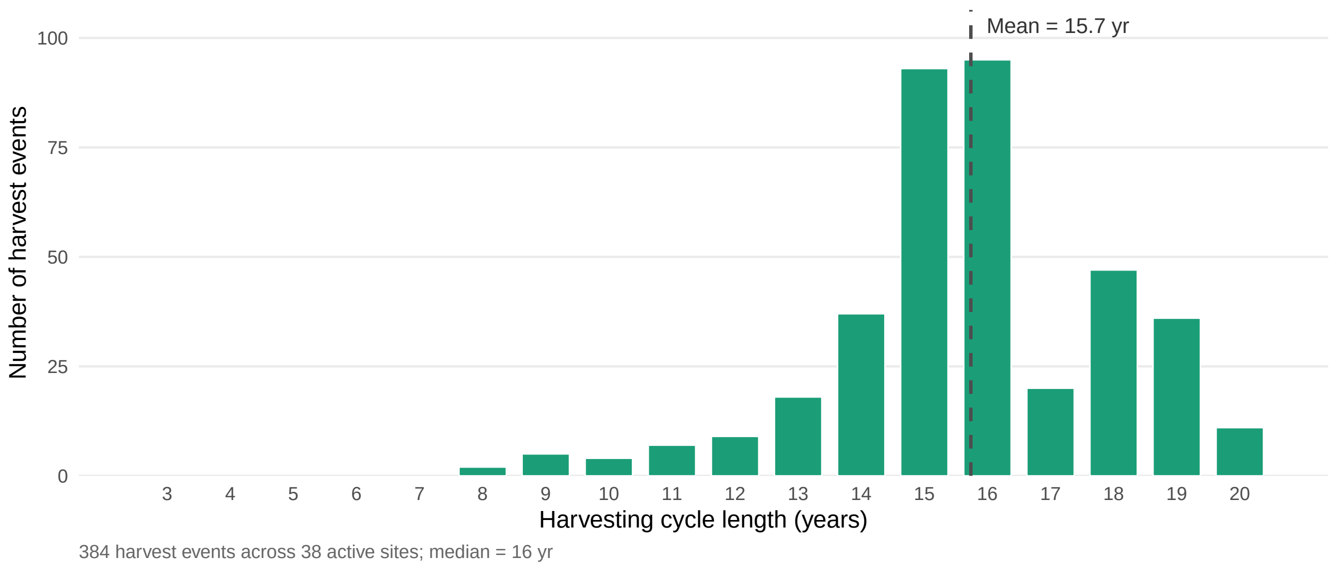


FIGURE 5 Distribution of harvesting cycle lengths at active agroforestry sites. Each bar represents the number of individual harvest events (over all periods and active sites) whose cycle length (stand age at harvest, i.e. $t - s$) falls in that year bin.

The histogram illustrates that majority of harvest events occur at stand ages between 14–18 years, indicating that long-rotation AFS seem to be more profitable in this setting. Short-rotation cycles (<10 years) occur only in marginal frequencies. This optimal rotation length might be caused either due to the limited demand of energy-grade biomass or the price gaps between biomass types or a combination of effects. The quantity of energy-grade biomass created in long-rotation AFS suffices to cover total energy-grade biomass demand almost completely. Rotation length is focused on production of chemical-grade biomass where highest prices are assumed. Focusing on chemical industry supply requires a production of trees with a comparatively large stems. This strategy maximizes revenue, but in the same time increases opportunity costs as wood harvests appear in larger time intervals. Figure 6 presents the aggregate cost–profit waterfall over all periods and actors.

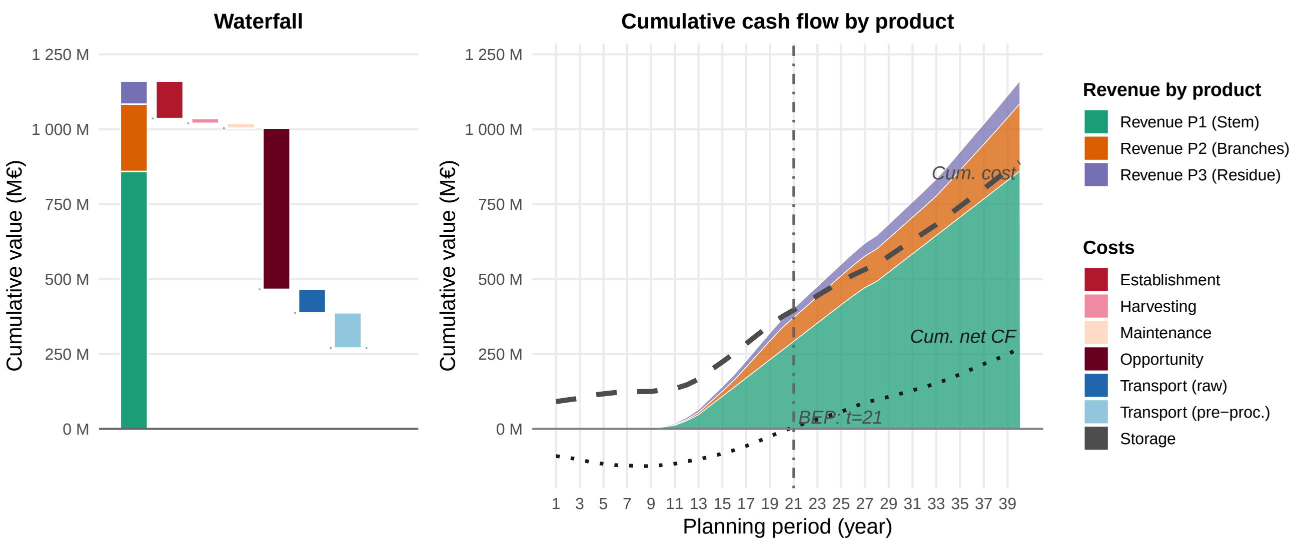


FIGURE 6 Total cost–profit breakdown across all actors and years. The left panel (waterfall) decomposes the objective value into revenue and individual cost categories; the right panel shows the period-level trajectory of revenues and stacked costs.

It appears that revenue is dominated by selling chemical-grade biomass and dominating cost component are opportunity cost although active sites are those sites with a comparatively small opportunity cost rate. Next to opportunity cost, establishment and transport costs are the most relevant cost drivers. Analyzing total cashflow over time shows that break-even is reached after about 20 years, i.e. after the first large-scale harvest occurred. Afterwards, cashflow is continuously increasing indicating that establishing AFS is a strategic change in agricultural practice and pays only if this strategy is implemented and maintained over a comparatively long time spans. This underpins the importance of strategic buying contracts with industrial partners in order to reduce revenue risks of farmers when producing AFS biomass.

Although profitability of the total AFS suply chain is given for the case study data, profitability of AFS operations on site-level are more heterogeneous. Figure 7 shows revenues and costs split by component and normalized to AFS size and active years of AFS.

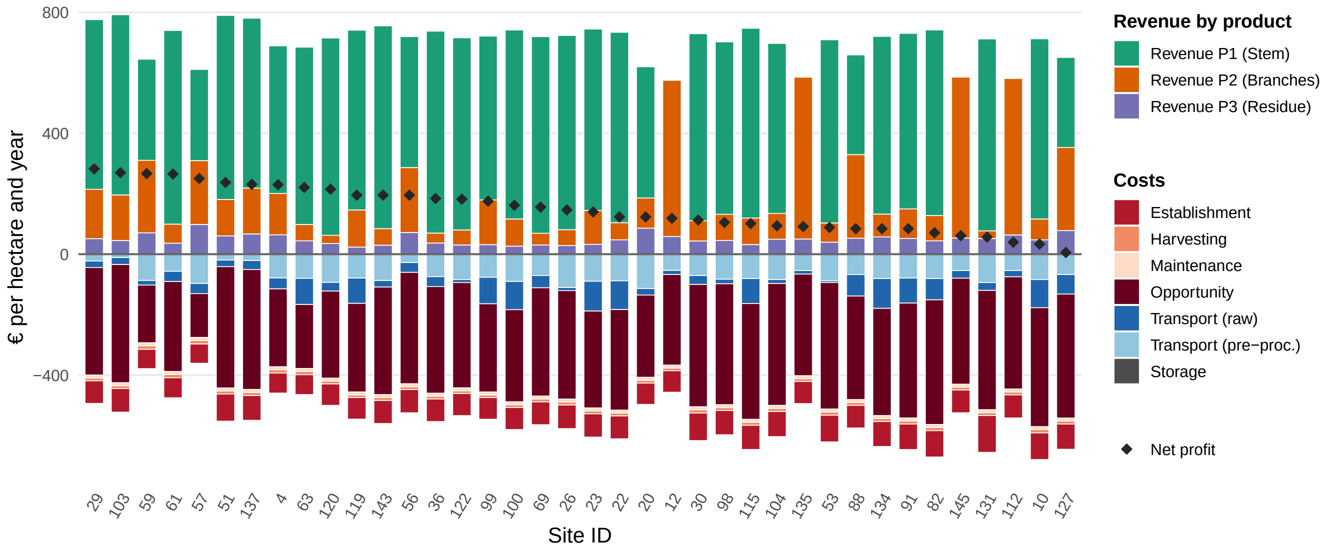


FIGURE 7 Revenues, costs and net profit for sites with AFS operations split by type and expressed in per hectare per year of AFS operation.

It appears that net profitability of AFS ranges between 282 and 6 € per hectare and active year. Note that this value constitutes additional profit after subtracting operational cost as well as imputed opportunity cost of lost profit from field crop. I.e., for calculating the realized net margin, opportunity can be added such that realized yearly net profit ranges between 473 and 210 € per hectare. Still, relative net profit is comparatively low compared to ordinary agricultural practices with profitable field crops (like winter wheat). The revenue splits of active sites indicates that at some sites medium-grade and low-grade biomass types contribute substantially or even solely to total revenue (see e.g. site 12, 135, 145, and 112). This indicates that site-specific AFS strategies might differ from the overall supply chain strategy. In other words, there seem to be strategic niches w.r.t. the consumer segments to be supplied. This might also be reflected in the rotation length and, although not discussed here, the selection of trees and clones for the AFS. Moreover, profitability of AFS sites seems to depend on various factors like local opportunity costs as well as closeness to consumers. Figure 8 shows net margins and opportunity cost rates of sites with active AFS.

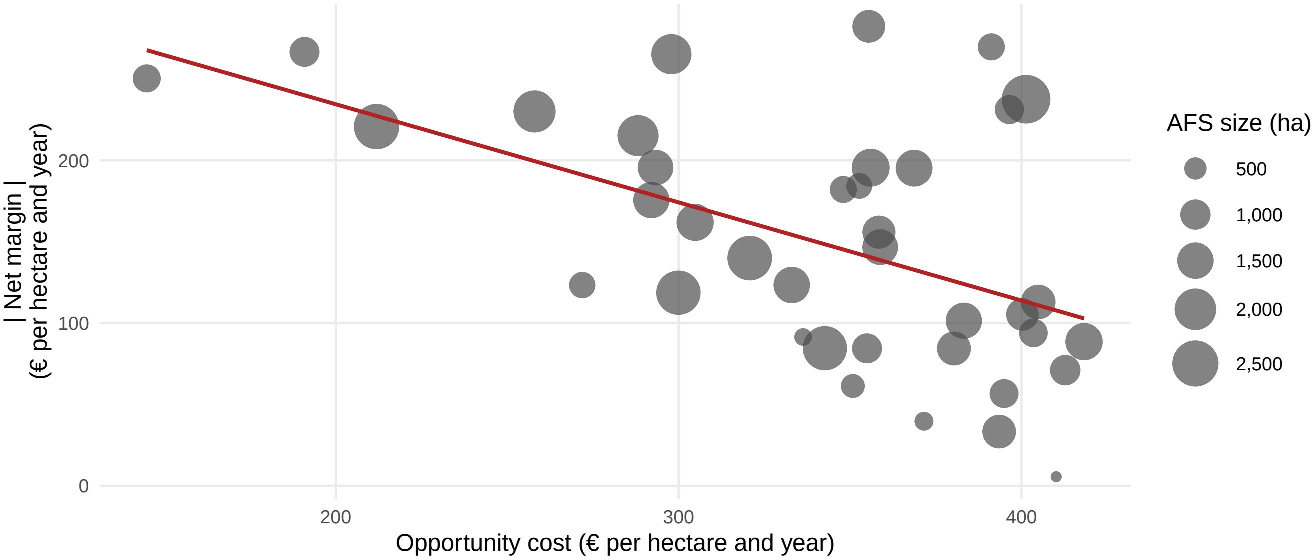


FIGURE 8 Scatterplot of opportunity cost rate and net margin (both in per ha and year) for each site with active AFS. Size of AFS reflected in point size.

As expected net margins of AFS depend heavily on a site's opportunity cost rate showing a clear negative trend. Moreover, it appears that AFS sizes vary quite heavily between 86 and 2794 hectare. Probably very small sites won't operate efficiently because cost for establishment, harvesting, and maintenance are assumed to scale with size, but are rather fixed below a certain critical threshold.

5 | CONCLUSION

This paper has developed and applied the AFS-SCD model, a MILP formulation for the integrated design and operation of agroforestry biomass supply chains. The model endogenises stand age dynamics through binary arc-based age-lag variables, couples component-specific Gompertz yield coefficients for stem, branch, and residue fractions with multi-echelon logistics decisions, and embeds a product quality cascade that allocates biomass grades to chemical, pulp, and energy markets within a unified optimisation framework. The formulation is, to the best of our knowledge, the first supply chain model to simultaneously optimise AFS establishment timing, consecutive harvest scheduling under rotation-age constraints, and product-grade allocation across spatially heterogeneous consumer markets.

The case study for the tri-state region of Saxony-Anhalt, northern Thuringia, and northern Saxony demonstrates the practical relevance of the approach. The optimal supply chain establishes AFS on only a subset of the available sites and area, concentrating establishment on parcels with comparatively low opportunity cost while sites with high opportunity cost are rarely selected. Long-rotation AFS strategies dominate the optimal solution, as they maximise the share of large-diameter stem biomass eligible for the highest-value chemical- and pulp-grade markets; revenue is correspondingly dominated by chemical-grade biomass sales, while opportunity cost, establishment, and transport costs are the most relevant cost drivers. Break-even at the aggregate supply chain level is reached after about 20 years, once the first large-scale harvest occurs, underscoring that AFS adoption constitutes a long-term strategic commitment that depends on reliable off-take agreements with industrial partners. At the site level, net profitability is highly heterogeneous and correlates negatively with local opportunity cost, indicating that AFS viability is strongly conditioned by regional land-use competition. These findings underscore the importance of product cascading as a strategic lever for AFS viability: maximising the share of stem biomass directed to high-value chemical and pulp applications substantially improves profitability compared to energy-only offtake strategies.

Methodologically, the AFS-SCD model bridges the gap between stand-level growth modelling and regional supply chain optimisation identified in the literature (Zahraee et al., 2020; Lamers et al., 2011). The age-lag constraint structure generalises to other perennial biomass crops with periodic coppice cycles, such as willow or black locust, and to long-rotation silvoarable systems in which rotation length is a strategic variable. The model is extensible to stochastic settings by replacing deterministic yield coefficients with scenario-weighted counterparts or by incorporating robust optimisation over yield uncertainty intervals (Sharma et al., 2013).

Several limitations point to directions for future research. First, consumer demand is treated as exogenous and time-invariant; endogenising demand responses to biomass price changes would require a partial-equilibrium extension. Second, the case study uses a single allometric scenario calibrated to Central European poplar; propagating Gompertz parameter uncertainty through to supply chain outcomes would strengthen the robustness analysis. Third, environmental co-benefits of AFS — including carbon sequestration, nitrate leaching reduction, and biodiversity enhancement — are not monetised in the current objective. Integrating life cycle assessment indicators as additional objectives or constraints would support a more comprehensive sustainability evaluation of AFS-based supply chains.

AUTHOR CONTRIBUTIONS

The author is responsible for all contents of this text.

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CONFLICT OF INTEREST

The author declares no potential conflict of interests.

DATA AVAILABILITY STATEMENT

The interactive web application used to parameterise, run, and explore the AFS-SCD model developed in this study, together with the underlying case study data, is openly accessible at `https://tk-mlu-pl.shinyapps.io/DIP_SmartAgroforst_SCD/`. All figures and results reported in this manuscript can be reproduced through this application.

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

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APPENDIX

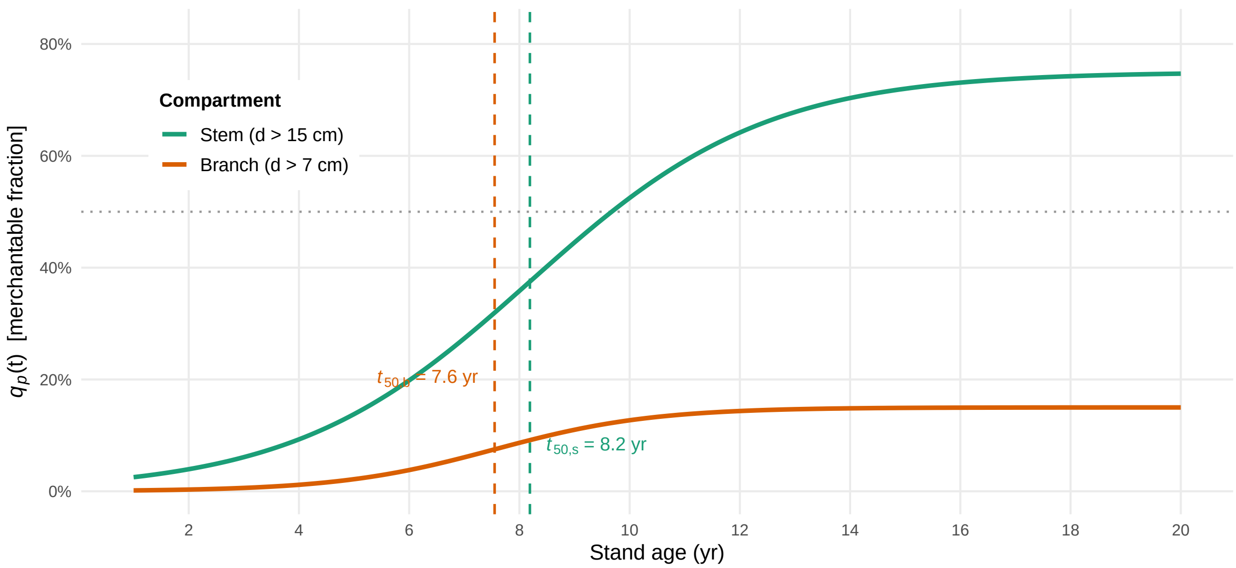

FIGURE 1 Logistic diameter-class merchantability functions $q_p(t)$ for stem (d > 15,cm, green) and branches (d > 7,cm, orange); the 50% threshold ages t_{50} are indicated by dashed vertical lines.

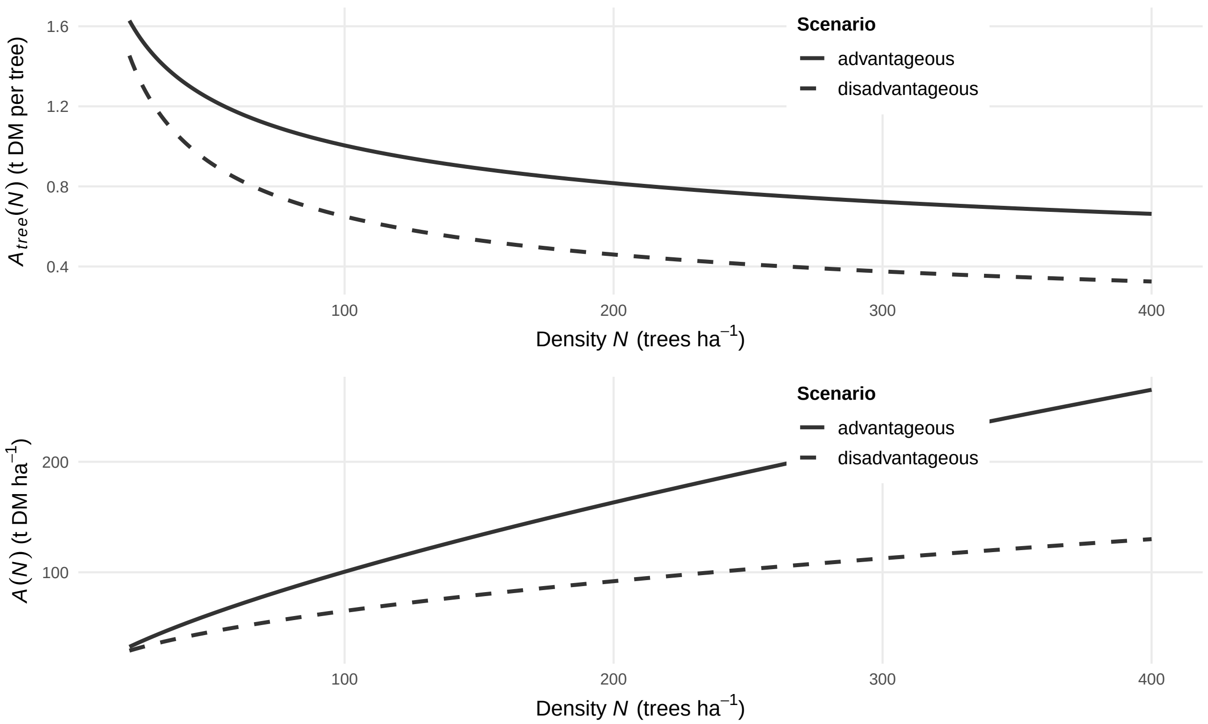

FIGURE 2 Tree-level and stand-level Gompertz asymptote functions $A_{tree}(N)$ and $A(N)$ for advantageous ($N = 4$, $\beta = -0.3$) and disadvantageous ($N = 6.5$, $\beta = -0.5$) site conditions.

TABLE 1 Notation for the biomass growth and harvest yield models.

Symbol	Description
$M(t)$	Total above- and belowground dry biomass per ha of AFS (t DM ha^{-1}) at stand age t
$A(N)$	Stand-level Gompertz asymptote (t DM ha^{-1}); density- and site-dependent
k	Gompertz growth-rate coefficient
t_0	Inflection age of stand biomass accumulation (years)
N	Stand density (trees ha^{-1})
C_{site}	Site-quality constant (kg ha$^{\beta-1}$); conservative: 4 475, good site: 6 471
β	Competition-density exponent; fitted $\hat{\beta} = 0.380$
$f_p(t)$	asymtotic fraction of AGB attributed to compartment $p \in \{s, b, r\}$ at age t
$q_p(t)$	Merchantable fraction of compartment p biomass at age t (logistic)
r_p	Logistic growth rate for merchantability of compartment p
$t_{50,p}$	Age at which 50% of compartment p biomass is merchantable (yr)
$\eta_p(t)$	Merchantable dry biomass of compartment p per ha at harvest age t (t DM ha^{-1})
ω	Moisture content at felling (mass fraction); $\omega = 0.55$

TABLE 2 Notation for the MILP model.

Sets & Indices	
$i \in \mathcal{I}$	Agroforestry sites
$j \in \mathcal{J}$	Storage / pre-treatment facilities
$k \in \mathcal{K}$	Consumer sites
$p \in \mathcal{P}$	Product types (stem, branches, residues)
$(p, p') \in \mathcal{Q}$	Admissible product-substitution pairs (quality cascade)
$\mathcal{T} = \{1, \ldots, T\}$	Planning periods (years)
$\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \ldots, T\}$	Feasible harvest periods
$\mathcal{S}^{\text{est}} = \{(0, t) \mid t \in \{1, \ldots, T - A^{\min}\}\}$	Establishment arcs: dummy source to period t
$\mathcal{S}^{\text{harv}} = \{(s, t) \mid s, t \in \mathcal{T},$ $t - s \in \{A^{\min}, \ldots, A^{\max}\}\}$	Harvest cycles within age-lag bounds
$\mathcal{S}^{\text{term}} = \{(s, T + 1) \mid$ $s \in \{A^{\min} + 1, \ldots, T\}\}$	Termination arcs to dummy sink $T + 1$
$\mathcal{S} = \mathcal{S}^{\text{est}} \cup \mathcal{S}^{\text{harv}} \cup \mathcal{S}^{\text{term}}$	Age-lag arc set (establishment ∪ harvest ∪ termination)
Decision Variables	
$z_{ist} \geq 0$	Area of site i with AFS operation between s and t
$X_{ijpt} \geq 0$	Transport flow (t) of p from site i to facility j in t
$X_{jkpp't} \geq 0$	Flow of p from facility j to consumer k satisfying demand p' in t
$S_{jpt} \geq 0$	Inventory (t) of product p at facility j end of period t
Parameters	
T	Length of planning horizon (years)
$A^{\min}, A^{\max}$	Minimum / maximum stand age at harvest (years)
AREA_i	Area of site i (ha)
η_{pa}	Yield coefficient (t/ha) of fraction p at stand age a, derived from (4)
R_{kp}	Revenue (/t) for product p at consumer k
C_i^{est}	Establishment cost (/ha) at site i
C_i^{opp}	Annual opportunity cost (/ha) at site i
C_i^{harv}	Harvest & refitting cost (/ha) at site i
$c^{\text{tr-raw}}$	Transport cost rate (/t km) for raw biomass
$c^{\text{tr-pre}}$	Transport cost rate (/t km) for pre-treated biomass
c_j^{stor}	Storage holding cost (/t) at facility j
d_{ij}, d_{jk}	Distance (km) between nodes
$\text{CAP}_j^{\text{stor}}/\text{CAP}_j^{\text{proc}}$	Storage and processing capacity (t) at facility j
$D_{kpt}^{\max}$	Maximum demand (t) of fraction p at consumer k in period t

TABLE 3 Consumer demand quantities and gate prices by product grade. P1: chemical/pulp; P2: sawlog/pulpwood; P3: energy/pellets. Demands in kt wood input per year; prices in /t wood input at the factory gate.

	Annual demand			Gate price		
Consumer	P1 (kt/yr)	P2 (kt/yr)	P3 (kt/yr)	P1 (/t)	P2 (/t)	P3 (/t)
consumer 1	300.0			50		
consumer 2	100.0			60		
consumer 3		300	50.0		37.5	24
consumer 4	200.0			47.5		
consumer 5		30	5.0		35	21
consumer 6		15	5.0		32.5	21
consumer 7		10	3.0		30	18
consumer 8		5	8.0		27.5	18
consumer 9	2.0	50	20.0	40	35	22.8
consumer 10			1.7			21
consumer 11			2.9			21
consumer 12			2.5			21
consumer 13			2.3			21
consumer 14			1.9			21
consumer 15			2.7			21
consumer 16			1.7			21
consumer 17			1.9			21
consumer 18			1.6			21
consumer 19			1.5			21
consumer 20			1.5			21
consumer 21			1.8			21
consumer 22			3.5			21
consumer 23			1.6			21
consumer 24			2.1			21
consumer 25			4.5			21
consumer 26			1.8			21
consumer 27			3.2			21
consumer 28	18.5			27.5		

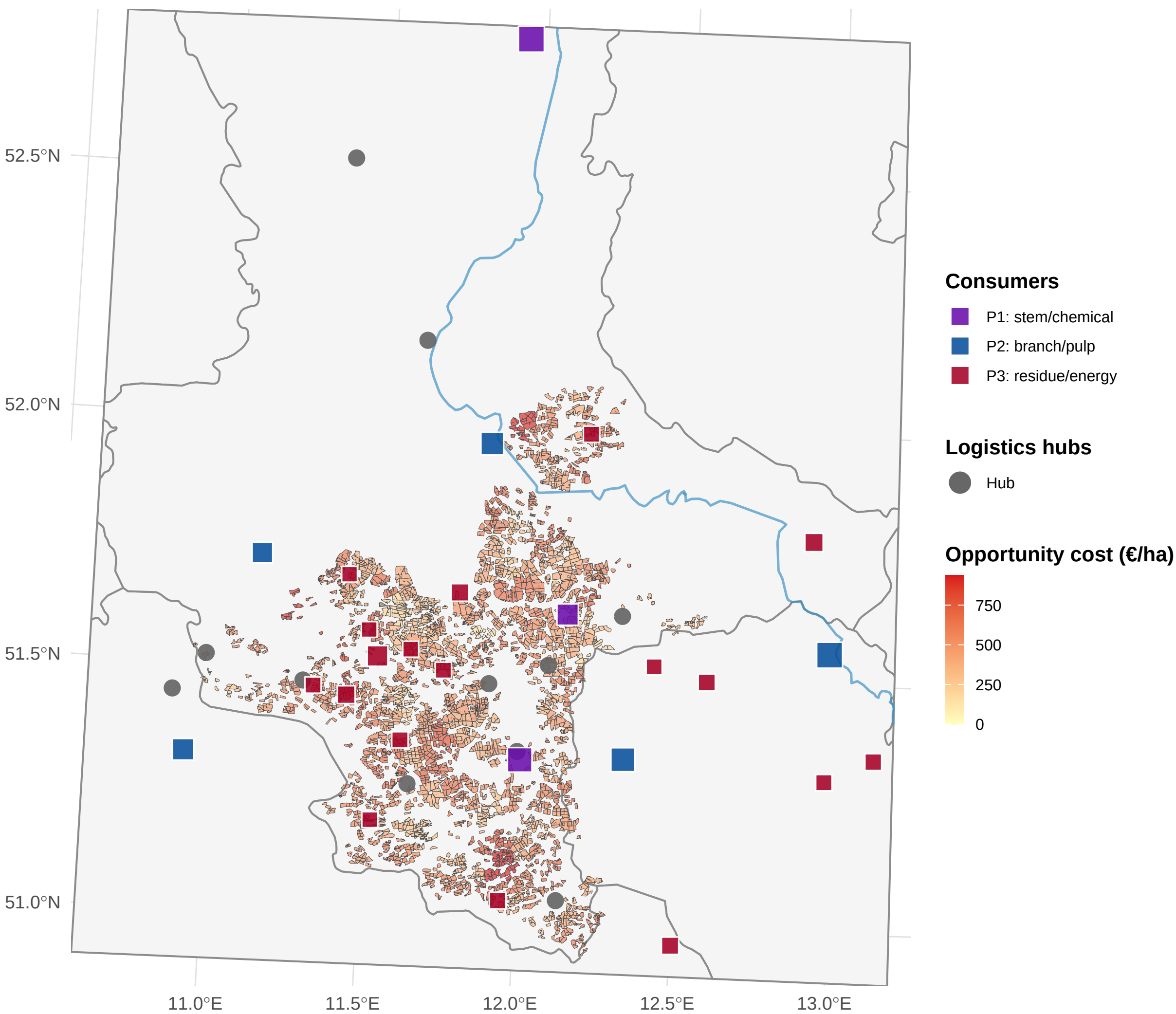

FIGURE 3 AFS supply chain network in Saxony-Anhalt. Grey lines: German federal state boundaries. Blue lines: major rivers. Coloured polygons: potential AFS parcels grouped by site (colours encode cluster membership; no legend). Orange symbols: pre-treatment facilities (circle = sawmill, triangle = logistics hub, size proportional to processing capacity). Coloured squares: consumer sites (violet: P1 chemical/pulp, blue: P2 pulp/paper, red: P3 energy/biogas, size proportional to total demand).

TABLE 4 Parameter values for the AFS-SCDcase study. Transport costs refer to road distance

Parameter	Description	Base-case value	Unit
Index sets			
$\lvert\mathcal{T}\rvert$	Planning periods	40	yr
$\lvert\mathcal{I}\rvert$	Sites	145	
$\lvert\mathcal{J}\rvert$	Pre-treatment / storage nodes	11	
$\lvert\mathcal{K}\rvert$	Consumers	28	
$\lvert\mathcal{P}\rvert$	Product grades	3	
Biomass dynamics			
$A^{\min}$	Minimum harvest age (years)	3	yr
$A^{\max}$	Maximum stand age (years)	20	yr
Cost parameters			
$c^{tr\text{-}raw}$	Raw transport cost	0.08	t^{-1} km^{-1}
$c^{tr\text{-}pre}$	Pre-processed transport cost	0.06	t^{-1} km^{-1}
c_i^{est}	Establishment cost rate	2500	ha^{-1} yr^{-1}
c_i^{harv}	harvest cost rate	150	ha^{-1} yr^{-1}
c_i^{main}	Maintenance cost rate	10	ha^{-1} yr^{-1}
c_i^{opp}	aver. opportunity cost	521	ha^{-1} yr^{-1}
Revenue parameters			
R_{P1}	Mean gate price P1	45	t^{-1}
R_{P2}	Mean gate price P2	33	t^{-1}
R_{P3}	Mean gate price P3	21	t^{-1}

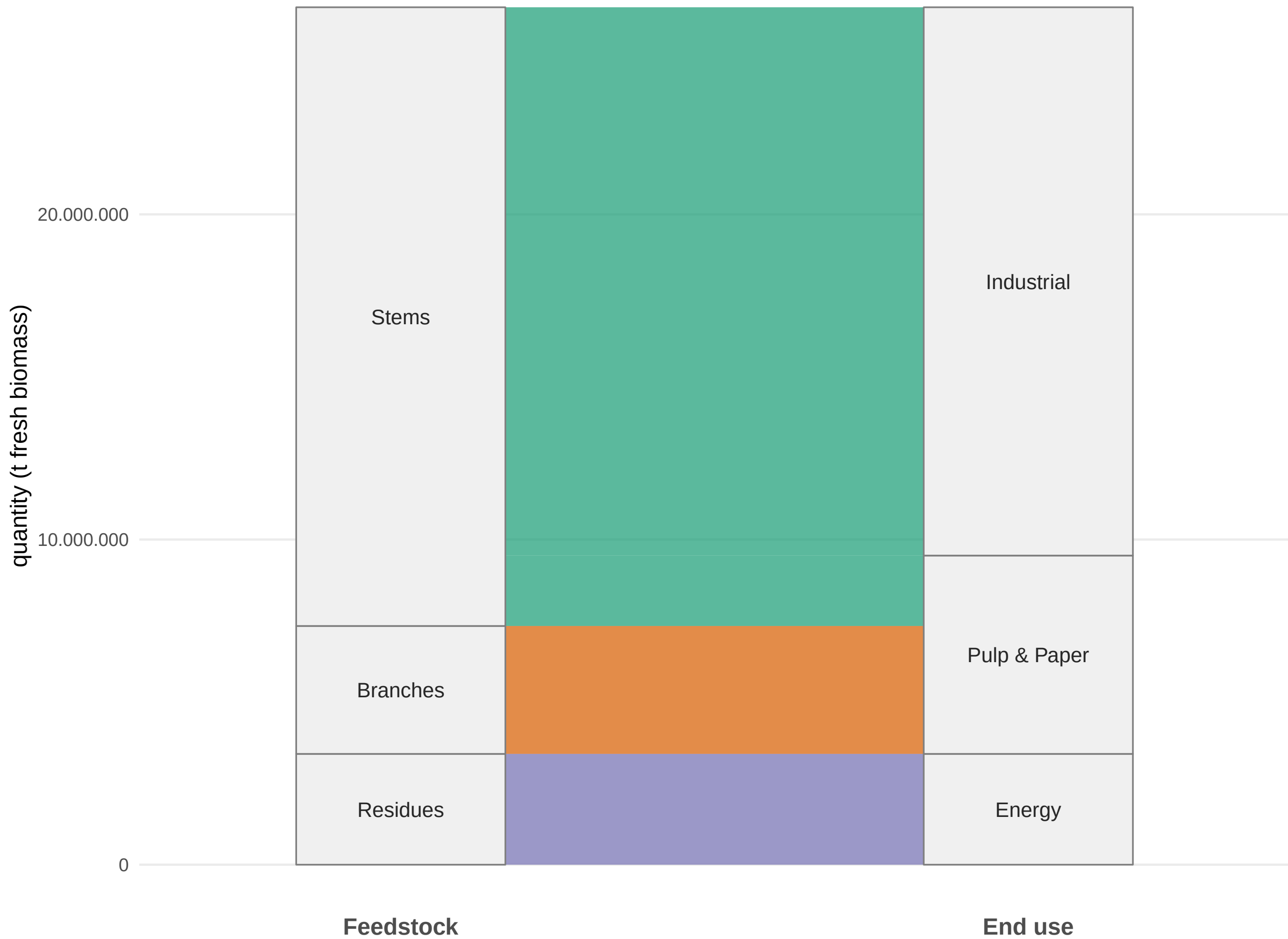

FIGURE 4 Total production quantity of biomass types and its industrial usage over planning horizon (in tons).